

DLX Factorial Program – Execution Trace

Initial Registers:

R1=0 R2=0 R3=0 R4=0 R5=0

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PC=000  LW    R1,0(R0)
          R1 = 3
          R2=0 R3=0 R4=0 R5=0

PC=001  ADDI  R2,R0,1
          R2 = 1
          R1=3 R3=0 R4=0 R5=0

PC=002  JAL   005
          R31 = 3

PC=005  SLEI  R3,R1,1
          R3 = 0
          R1=3 R2=1 R4=0 R5=0

PC=006  BNEZ  R3,00F
          R3=0 → branch not taken

PC=007  ADDI  R4,R0,0
          R4 = 0

PC=008  ADDI  R5,R1,0
          R5 = 3

PC=009  ADD   R4,R4,R2
          R4 = 1

PC=00A  SUBI  R5,R5,1
          R5 = 2

PC=00B  BNEZ  R5,009
          R5=2 → branch taken

PC=009  ADD   R4,R4,R2
          R4 = 2

PC=00A  SUBI  R5,R5,1
          R5 = 1

PC=00B  BNEZ  R5,009
          R5=1 → branch taken

PC=009  ADD   R4,R4,R2
          R4 = 3

PC=00A  SUBI  R5,R5,1
          R5 = 0

PC=00B  BNEZ  R5,009
          R5=0 → branch not taken

PC=00C  ADDI  R2,R4,0
          R2 = 3

PC=00D  SUBI  R1,R1,1
          R1 = 2

PC=00E  J     005

PC=005  SLEI  R3,R1,1
          R3 = 0
          R1=2 R2=3

PC=006  BNEZ  R3,00F
```

R3=0 → branch not taken

PC=007 ADDI R4,R0,0
R4 = 0

PC=008 ADDI R5,R1,0
R5 = 2

PC=009 ADD R4,R4,R2
R4 = 3

PC=00A SUBI R5,R5,1
R5 = 1

PC=00B BNEZ R5,009
R5=1 → branch taken

PC=009 ADD R4,R4,R2
R4 = 6

PC=00A SUBI R5,R5,1
R5 = 0

PC=00B BNEZ R5,009
R5=0 → branch not taken

PC=00C ADDI R2,R4,0
R2 = 6

PC=00D SUBI R1,R1,1
R1 = 1

PC=00E J 005

PC=005 SLEI R3,R1,1
R3 = 1

PC=006 BNEZ R3,00F
branch taken

PC=00F JR R31
PC = 3

PC=003 SW 1(R0),R2
MEM[1] = 6

PC=004 J 004

Final Registers:

R1 = 1

R2 = 6

R3 = 1

R4 = 6

R5 = 0

Memory:

mem[1] = 6